

Reproducibility Report: Maglio & Polman (2014), Study 1

Minimal version

Mikka Hoffman

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Goals

1. Load a published dataset with `here()` and `read_excel()`
2. Reproduce the ANOVA and effect sizes reported in the paper
3. Say clearly what did and did not reproduce, and why

This is the **minimal** version: the structure and the questions are here, and you write the code. If you get stuck, `reproducibility-report-bumper-rails.qmd` has the same steps with most of the code filled in. It is fine to peek at it for one step and then come back.

error: `true` is set at the top of this file, so the document still renders while some chunks are unfinished. The chunks that still fail show their error message in red.

Before you start, read Study 1 in the paper: [original_paper/maglio-polman-2014.pdf](#) (two pages).

The study

Maglio, S. J., & Polman, E. (2014). Spatial orientation shrinks and expands psychological distance. *Psychological Science*, 25(7), 1345–1352. <https://doi.org/10.1177/0956797614530571>

Methods summary. Researchers recruited 202 volunteers at the Bay Street subway station in Toronto. Half were waiting on the westbound platform and half on the eastbound platform. Each participant was randomly assigned to rate how far away one of four stations felt (1 = very close, 7 = very far): Spadina (two stops west), St. George (one stop west), Bloor-Yonge (one stop east), or Sherbourne (two stops east). The authors ran a 2×4 ANOVA on the distance ratings, then compared eastbound and westbound travelers separately at each station.

Target outcomes. This is the result you are trying to reproduce, quoted from the Study 1 results section:

We carried out a 2 (orientation: toward, away from) $\times$ 4 (station: Spadina, St. George, Bloor-Yonge, Sherbourne) analysis of variance (ANOVA) on closeness ratings, which revealed no main effect of orientation, $F < 1$, and a main effect of station, $F(3, 194) = 24.10$, $p < .001$, $p^2 = .27$. This main effect was qualified by the predicted interaction between orientation and station, $F(3, 194) = 16.28$, $p < .001$, $p^2 = .20$. We decomposed this interaction by the subjective-distance ratings between participants traveling east and west for each of the four subway stations. Westbound participants rated the stations to the west of Bay Street as closer than did eastbound participants; this effect was obtained for both the station one stop to the west (St. George, $p < .001$, $p^2 = .28$) and the station two stops to the west (Spadina, $p = .001$, $p^2 = .20$). The opposite pattern held true for stations to the east of Bay Street. Eastbound participants rated the stations to the east of Bay Street as closer than did westbound participants; this effect was obtained for both the station one stop to the east (Bloor-Yonge, $p = .053$, $p^2 = .08$) and the station two stops to the east (Sherbourne, $p < .001$, $p^2 = .24$). Figure 1 summarizes these results.

p^2 is **partial eta squared**, an effect size. You will compute it below.

Step 1: Load packages

`readxl` (reads Excel files) and `broom` (turns model output into tibbles) are installed along with the tidyverse, but `library(tidyverse)` does not load them, so we load them by name.

```
install.packages(c("tidyverse", "here"))
```

```
library(tidyverse)
library(readxl)
library(broom)
library(here)
```

Step 2: Load the data

The authors posted one Excel file per study. We only need Study 1. The columns are described in `docs/codebook.md`.

```
d <- read_excel(here("data", "S1_Subway.xlsx"))

glimpse(d)
```

Rows: 202

Columns: 4

```
$ DIRECTION <chr> "EAST", "EAST", "EAST", "EAST", "EAST", "EAST", "EAST", "EA~
$ DISTANCE  <dbl> 1, 2, 2, 3, 1, 1, 2, 1, 3, 2, 1, 1, 2, 1, 3, 1, 2, 1, 1, 1,~
$ STN_NUMBER <dbl> 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3,~
$ STN_NAME  <chr> "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-Y", "B-
Y", "B--
```

Step 3: Tidy the data

The data are already tidy: one row per participant, one column per variable. But R has read `DIRECTION` and `STN_NAME` in as text. For an ANOVA they need to be **factors**, and we set the order of the station levels to run west to east, so tables and plots come out in map order.

Use `mutate()` and `factor()` to convert both columns. For `STN_NAME`, give `factor()` a `levels` argument listing the stations from west to east.

```
# convert direction and station name to factors
d <- d |>
  mutate(DIRECTION = factor(DIRECTION),
         STN_NAME = factor(STN_NAME, levels = c("SPAD", "STG", "B-Y", "SHER")))
```

Step 4: Reproduce the analysis

How many people are in each cell?

Count the participants in each direction $\times$ station combination. Does it match the paper (202 total, 101 per platform)? Is every cell the same size?

```
d_count <- d |>
  count(DIRECTION, STN_NAME) |>
  pivot_wider(names_from = STN_NAME, values_from = n)
print(d_count)
```

```
# A tibble: 2 x 5
  DIRECTION SPAD  STG `B-Y`  SHER
  <fct>      <int> <int> <int> <int>
1 EAST         26   26   23   26
2 WEST         25   25   26   25
```

The 2 $\times$ 4 ANOVA

Fit a two-way ANOVA with `aov()`: distance predicted by direction, station, and their interaction. Remember that `y ~ A * B` is shorthand for `y ~ A + B + A:B`.

```
anova <- aov(DISTANCE ~ DIRECTION * STN_NAME, data = d)
summary(anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
DIRECTION	1	0.71	0.713	0.664	0.416
STN_NAME	3	75.16	25.053	23.349	6.01e-13 ***
DIRECTION:STN_NAME	3	52.41	17.471	16.283	1.77e-09 ***
Residuals	194	208.15	1.073		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

The numbers in parentheses in “F(3, 194)” are the degrees of freedom: 3 for the effect, 194 for the residuals. Check that yours match.

The teaching team gets:

- Main effect of direction: $F(1, 194) = 0.66$
- Main effect of station: $F(3, 194) = 23.35$

- Interaction between direction and station: $F(3, 194) = 16.28$

Two of those match the paper. One is close but not the same. Keep going; the bonus question at the end asks you to work out why.

Partial eta squared

Partial eta squared is the share of variance an effect accounts for, once the other effects are set aside:

$$\eta_p^2 = \frac{SS_{\text{effect}}}{SS_{\text{effect}} + SS_{\text{residual}}}$$

Compute it for each effect. Hint: `tidy()` from `broom` turns the ANOVA table into a tibble, with the sums of squares in a column called `sumsq`.

```
#anova table
anova_table <- tidy(anova)
anova_table
```

```
# A tibble: 4 x 6
  term          df  sumsq meansq statistic  p.value
<chr>      <dbl> <dbl> <dbl>    <dbl>    <dbl>
1 DIRECTION      1  0.713  0.713    0.664 4.16e- 1
2 STN_NAME       3 75.2    25.1    23.3 6.01e-13
3 DIRECTION:STN_NAME 3 52.4    17.5    16.3 1.77e- 9
4 Residuals    194 208.     1.07    NA    NA
```

```
# calculate partial eta squared
ss_residual <- anova_table |>
  filter(term == "Residuals") |>
  pull(sumsq)

anova_table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq/(sumsq + ss_residual)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 3 x 5
  term          df statistic  p.value partial_eta_sq
<chr>      <dbl>    <dbl>    <dbl>    <dbl>
```

1	DIRECTION	1	0.664	4.16e- 1	0.00341
2	STN_NAME	3	23.3	6.01e-13	0.265
3	DIRECTION:STN_NAME	3	16.3	1.77e- 9	0.201

Do your values round to the paper's .27 (station) and .20 (interaction)?

Station-by-station comparisons

Now compare eastbound and westbound travelers at each station separately. For each station: keep only that station's rows, compute the mean rating in each direction, run a one-way ANOVA of distance on direction, and compute p^2 .

St. George

```
#select and summarize only stg
stg <- d |>
  filter(STN_NAME == "STG")
stg |>
  group_by(DIRECTION) |>
  summarise(mean = mean(DISTANCE),
            n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean    n
  <fct>      <dbl> <int>
1 EAST      2.77    26
2 WEST      1.64    25
```

```
#anova
stg.anova <- aov(DISTANCE ~ DIRECTION, data = stg)
stg.anova
```

Call:

```
aov(formula = DISTANCE ~ DIRECTION, data = stg)
```

Terms:

	DIRECTION	Residuals
Sum of Squares	16.25207	42.37538
Deg. of Freedom	1	49

Residual standard error: 0.9299483
Estimated effects may be unbalanced

```
stg.anova.table <- tidy(stg.anova)

#partial eta squared
stg_ss_residual <- stg.anova.table |>
  filter(term == "Residuals") |>
  pull(sumsq)

stg.anova.table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq/(sumsq + stg_ss_residual)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 1 x 5
  term      df statistic  p.value partial_eta_sq
<chr>   <dbl>   <dbl>   <dbl>         <dbl>
1 DIRECTION     1     18.8 0.0000723         0.277
```

Spadina

```
#select and summarize only spad
spad <- d |>
  filter(STN_NAME == "SPAD")
spad |>
  group_by(DIRECTION) |>
  summarise(mean = mean(DISTANCE),
            n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean    n
<fct>     <dbl> <int>
1 EAST      3.65    26
2 WEST      2.64    25
```

```
#anova
spad.anova <- aov(DISTANCE ~ DIRECTION, data = spad)
spad.anova
```

Call:

```
aov(formula = DISTANCE ~ DIRECTION, data = spad)
```

Terms:

	DIRECTION	Residuals
Sum of Squares	13.10048	53.64462
Deg. of Freedom	1	49

Residual standard error: 1.046321

Estimated effects may be unbalanced

```
spad.anova.table <- tidy(spad.anova)

#partial eta squared
spad_ss_residual <- spad.anova.table |>
  filter(term == "Residuals") |>
  pull(sumsq)

spad.anova.table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq/(sumsq + spad_ss_residual)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

A tibble: 1 x 5

	term	df	statistic	p.value	partial_eta_sq
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	DIRECTION	1	12.0	0.00113	0.196

Bloor-Yonge

```
#select and summarize only b-y
b_y <- d |>
  filter(STN_NAME == "B-Y")
b_y |>
  group_by(DIRECTION) |>
  summarise(mean = mean(DISTANCE),
            n = n())
```

A tibble: 2 x 3

DIRECTION	mean	n
<fct>	<dbl>	<int>

1	EAST	1.61	23
2	WEST	2.19	26

```
#anova
b_y.anova <- aov(DISTANCE ~ DIRECTION, data = b_y)
b_y.anova
```

Call:

```
aov(formula = DISTANCE ~ DIRECTION, data = b_y)
```

Terms:

	DIRECTION	Residuals
Sum of Squares	4.15675	49.51672
Deg. of Freedom	1	47

Residual standard error: 1.026425

Estimated effects may be unbalanced

```
b_y.anova.table <- tidy(b_y.anova)

#partial eta squared
b_y_ss_residual <- b_y.anova.table |>
  filter(term == "Residuals") |>
  pull(sumsq)

b_y.anova.table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq/(sumsq + b_y_ss_residual)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

A tibble: 1 x 5

	term	df	statistic	p.value	partial_eta_sq
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1	DIRECTION	1	3.95	0.0528	0.0774

Sherbourne

```
#select and summarize only sher
sher <- d |>
  filter(STN_NAME == "SHER")
```

```
sher |>
  group_by(DIRECTION) |>
  summarise(mean = mean(DISTANCE),
            n = n())
```

```
# A tibble: 2 x 3
  DIRECTION mean    n
  <fct>     <dbl> <int>
1 EAST      2.77    26
2 WEST      4      25
```

```
#anova
sher.anova <- aov(DISTANCE ~ DIRECTION, data = sher)
sher.anova
```

Call:

```
aov(formula = DISTANCE ~ DIRECTION, data = sher)
```

Terms:

	DIRECTION	Residuals
Sum of Squares	19.30618	62.61538
Deg. of Freedom	1	49

Residual standard error: 1.130427

Estimated effects may be unbalanced

```
sher.anova.table <- tidy(sher.anova)

#partial eta squared
sher_ss_residual <- sher.anova.table |>
  filter(term == "Residuals") |>
  pull(sumsq)

sher.anova.table |>
  filter(term != "Residuals") |>
  mutate(partial_eta_sq = sumsq/(sumsq + sher_ss_residual)) |>
  select(term, df, statistic, p.value, partial_eta_sq)
```

```
# A tibble: 1 x 5
  term      df statistic p.value partial_eta_sq
  <chr>   <dbl>   <dbl>   <dbl>         <dbl>
1 DIRECTION 1      15.1 0.000305      0.236
```

Fill in this table as you go:

Station	Paper p	Your p	Paper p^2	Your p^2
St. George	< .001	< .001	.28	0.28
Spadina	.001	.001	.20	.20
Bloor-Yonge	.053	.053	.08	.08
Sherbourne	< .001	< .001	.24	.024

Step 5: Reflection

Were you able to reproduce the results you attempted to reproduce? If not, which parts could you not reproduce?

I was able to reproduce all of the results except for the main effect of station.

How difficult was it to reproduce the results?

It was fairly easy to reproduce the results, especially with this scaffolding.

What made it hard, and what made it easy? Think about the data files, the variable names, and how much the paper told you about the analysis.

Clear and easy variable names, clean and streamlined data files, and clearly specified statistical tests helped to make reproducing results easier. Having to do some amount of data cleaning/transformation (e.g., having extra spaces in some cells, having to recode as factors) made it a bit harder, and not having available data scripts or clear analysis methods in their paper (e.g., they don't say if they analyzed data in R or SPSS) made it harder.

Bonus questions

1. Why is the station F different? Your F for station (23.35) is close to the paper's (24.10) but not equal, while the interaction matches exactly. Try refitting the model with the predictors in the opposite order, `DISTANCE ~ STN_NAME * DIRECTION`. Does anything change? Look back at your cell counts. What does that suggest about how the authors' software (SPSS) and R's `aov()` each divide up the variance? `docs/anova-in-r.md` explains it, but try first.

```
anova <- aov(DISTANCE ~ STN_NAME * DIRECTION, data = d)
summary(anova)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
STN_NAME	3	75.47	25.156	23.446	5.41e-13 ***
DIRECTION	1	0.40	0.402	0.375	0.541
STN_NAME:DIRECTION	3	52.41	17.471	16.283	1.77e-09 ***
Residuals	194	208.15	1.073		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

2. Orientation or direction? The paper describes the factor as “orientation: toward, away from”. The dataset has DIRECTION: east or west. Are those the same variable? If not, which one did the authors actually put in their ANOVA? Make an `orientation` column with `if_else()` and refit the model with it to check.

```
d <- d |>
  mutate(orientation = if_else(DIRECTION == "EAST" & STN_NUMBER>2, "toward", if_else(DIRECTION == "WEST", "away", "toward")))
orientation_anova <- aov(DISTANCE ~ STN_NAME * orientation, data = d)
tidy(orientation_anova)
```

A tibble: 4 x 6

	term	df	sumsq	meansq	statistic	p.value
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	STN_NAME	3	75.5	25.2	21.9	3.02e-12
2	orientation	1	33.9	33.9	29.4	1.71e- 7
3	STN_NAME:orientation	2	2.69	1.35	1.17	3.12e- 1
4	Residuals	195	224.	1.15	NA	NA

3. Reproduce Figure 1. Plot the mean distance rating for each station, split by direction, with error bars of ± 1 standard error, as in the paper. `stat_summary(fun.data = mean_se)` will draw both the means and the error bars for you.

```
d_summary <- d |>
  group_by(DIRECTION) |>
  summarise(mean = mean(DISTANCE))
ggplot(data = d_summary, aes(x=STN_NAME, y = mean, color = DIRECTION, group = DIRECTION))+
  geom_point()+
  geom_line()
```

```
Error in `geom_point()`:
! Problem while computing aesthetics.
! Error occurred in the 1st layer.
Caused by error:
! object 'STN_NAME' not found
```

Going further

Instead of four near-identical chunks, do all four stations in one pipeline. Look up `nest()` and `map()`.